

Polar Unit Test (Practice F)

Calculator

You have the entire period.

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this test. _____

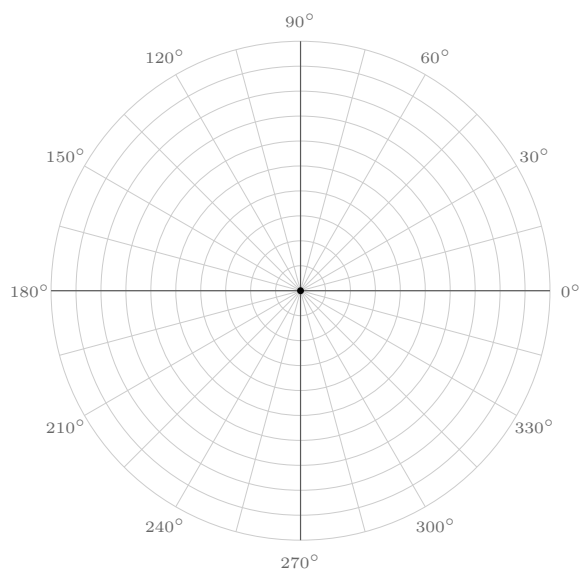
SHOW ALL WORK FOR FULL CREDIT.

Signature _____

Use your calculator to sketch a graph of each polar equation. Identify the type of polar graph it is. Show your set up (what you are putting into the calculator). (3 points each)

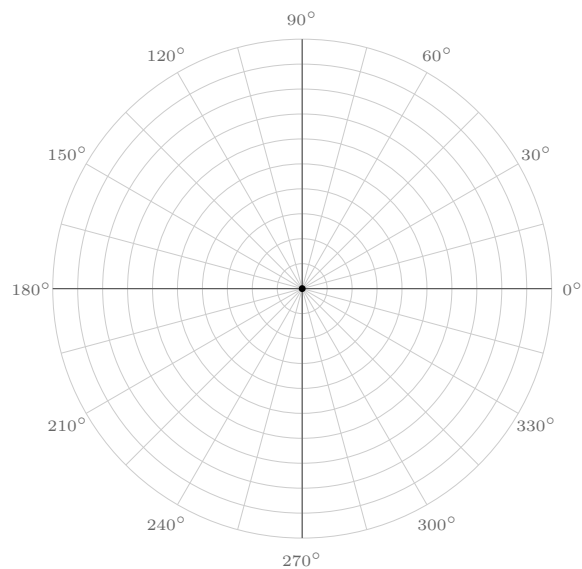
Word bank: circle cardioid limaçon rose curve lemniscate

1. $r = 3 + 3 \sin \theta$



Type: _____

2. $r = 5 \cos 3\theta$



Type: _____

Polar Unit Test (Practice F)

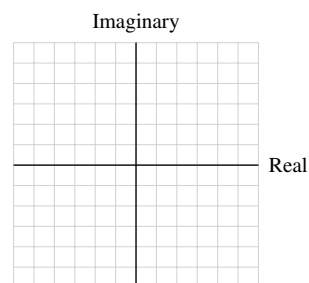
Non-Calculator

SHOW ALL WORK FOR FULL CREDIT.

Simplify. (3 points) 3. $\frac{\sqrt{10}}{\sqrt{-5} \cdot \sqrt{-8}}$	Simplify. (2 points) 4. $\frac{-9 + \sqrt{-18}}{6}$
Perform the indicated operation. Write the answer in rectangular form. (2 points) 5. i^{-22}	Perform the indicated operation. Write the answer in rectangular form. (2 points) 6. $(-7 + 3i) - (2 - 5i) - (-4 + 9i)$
Perform the indicated operation. Write the answer in rectangular form. (3 points) 7. $\frac{5 - i}{2 + 3i}$	Perform the indicated operation. Write the answer in rectangular form. (3 points) 8. $-i(3 + 2i)(1 - 4i)$
Solve over the set of complex numbers. (3 points) 9. $x^2 - 6x + 25 = 0$	Convert to trig form. (3 points) 10. $-5\sqrt{3} - 5i$

Find the sum. Write it in standard form. Then graph both complex numbers and their resultant, and find the absolute value of the sum. (3 points)

11. $(-4 + i) + (6 + 3i)$



Using $z^2 - 1, (z^2 - 1)^2 - 1, [(z^2 - 1)^2 - 1]^2 - 1, \dots$ is the following in the Julia set? Show ALL work. (3 points)

12. $z = -1$

YES / NO

Perform the indicated operation. Write each answer in rectangular form. (3 points each)

13. $\frac{12(\cos 100^\circ + i \sin 100^\circ)}{4(\cos 340^\circ + i \sin 340^\circ)}$

14. $\left[5 \operatorname{cis} \frac{5\pi}{6} \right] \left[2 \operatorname{cis} \frac{3\pi}{2} \right]$

Using $z, z^2 + z, (z^2 + z)^2 + z, \dots$ is the following number in the Mandelbrot set? Show ALL work. (3 points)

15. $z = 1$

YES / NO

Find each power. Write your answer in rectangular form. (3 pts)

16. $(\sqrt{2} + i\sqrt{2})^{10}$

Find all complex number solutions of each equation. Leave answers in trigonometric form. (4 pts)

17. $x^4 - 81i = 0$

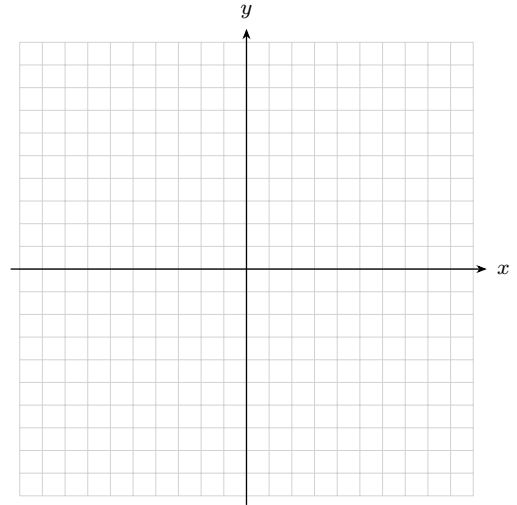
Give two other pairs of polar coordinates for the point: one with $r > 0$ and one with $r < 0$, with $-2\pi \leq \theta < 2\pi$. (3 points) 18. $(-3, \frac{4\pi}{3})$	Convert the polar coordinates to rectangular coordinates. Write your answer in exact form. (3 points) 19. $(-8, \frac{2\pi}{3})$
Convert the rectangular equation to a polar equation. Solve for r. (3 points) 20. $x^2 + y^2 = -10y$	Find the polar equation of the line that passes through the two points. Simplify your answer. (3 points) 21. $(4, 180^\circ)$ and $(2\sqrt{2}, 135^\circ)$

Convert each polar equation to a rectangular equation and graph it. (No calculator) (4 points each)

22. $r = \frac{6}{1 - \cos \theta}$

Convert each polar equation to a rectangular equation and graph it. (No calculator) (4 points each)

23. $r = 6 \cos \theta - 8 \sin \theta$



Find all points of intersection for $0 \leq \theta < 2\pi$. Include the pole if the curves share it. (4 points)

24. $r = 3 \sin \theta$, $r = 3 \cos \theta$

Identify the type of polar graph. Use the word bank: circle, cardioid, limaçon, rose curve, lemniscate. (2 points)

25. $r = 6 - 2 \cos \theta$

Type: _____

ANSWER KEY – Calculator

Left: the work, one step per line, the way it goes on paper. Right: why that line happens.

Pace check: about 50 minutes for a strong pass. Calculator part about 5, #3–12 about 15, #13–17 about 13, #18–25 about 15. That leaves about 30 minutes of the period to check your work.

Use your calculator to sketch a graph of the polar equation. Identify the type of polar graph it is. (3 points)

1. $r = 3 + 3 \sin \theta$

WORK

Polar mode: $r1(\theta) = 3 + 3 \sin \theta$

$90^\circ: r = 6$; $270^\circ: r = 0$; $0^\circ, 180^\circ: r = 3$

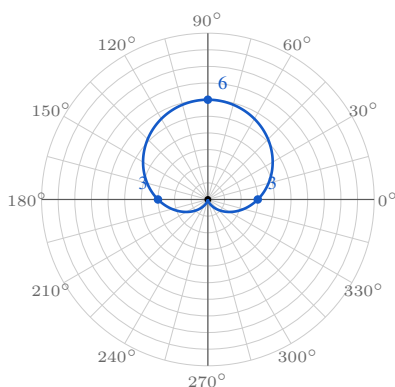
$$\frac{a}{b} = 1$$

cardioid

WHY

Degree mode needs θ from 0 to 360 (radians: 0 to 2π).

Cusp at the pole.



Use your calculator to sketch a graph of the polar equation. Identify the type of polar graph it is. (3 points)

2. $r = 5 \cos 3\theta$

WORK

Polar mode: $r1(\theta) = 5 \cos(3\theta)$

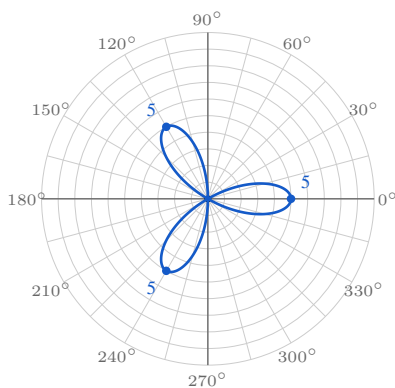
$n = 3$ odd: 3 petals, each 5 long

tips at $0^\circ, 120^\circ, 240^\circ$

rose curve

WHY

$\cos$ puts one on the x -axis.



ANSWER KEY – Non-Calculator

Left: the work, one step per line, the way it goes on paper. Right: why that line happens.

Simplify. (3 points)

$$3. \frac{\sqrt{10}}{\sqrt{-5} \cdot \sqrt{-8}}$$

WORK

$$\frac{\sqrt{10}}{i^2 \sqrt{40}} = \frac{\sqrt{10}}{-2\sqrt{10}}$$

$$\boxed{-\frac{1}{2}}$$

WHY

Pull out i first.

Simplify. (2 points)

$$4. \frac{-9 + \sqrt{-18}}{6}$$

WORK

$$\sqrt{-18} = 3i\sqrt{2}$$

$$\boxed{-\frac{3}{2} + \frac{\sqrt{2}}{2}i}$$

WHY

Reduce each part by 3.

Perform the indicated operation. Write the answer in rectangular form. (2 points)

$$5. i^{-22}$$

WORK

$$-22 + 24 = 2$$

$$i^2 = \boxed{-1 + 0i}$$

WHY

Add a multiple of 4.

Perform the indicated operation. Write the answer in rectangular form. (2 points)

$$6. (-7 + 3i) - (2 - 5i) - (-4 + 9i)$$

WORK

$$-7 + 3i - 2 + 5i + 4 - 9i$$

$$\boxed{-5 - i}$$

WHY

Perform the indicated operation. Write the answer in rectangular form. (3 points)

7. $\frac{5-i}{2+3i}$

WORK

$$\frac{2-3i}{2-3i}$$

top: $10 - 15i - 2i + 3i^2 = 7 - 17i$

bottom: $4 + 9 = 13$

$$\frac{7}{13} - \frac{17}{13}i$$

WHY

$$3i^2 = -3.$$

Perform the indicated operation. Write the answer in rectangular form. (3 points)

8. $-i(3+2i)(1-4i)$

WORK

$$(3+2i)(1-4i) = 3 - 12i + 2i - 8i^2 = 11 - 10i$$

$$-i(11 - 10i) = -11i + 10i^2 = \boxed{-10 - 11i}$$

WHY

Solve over the set of complex numbers. (3 points)

9. $x^2 - 6x + 25 = 0$

WORK

$$x = \frac{6 \pm \sqrt{36 - 100}}{2} = \frac{6 \pm 8i}{2}$$

$$\boxed{x = 3 \pm 4i}$$

WHY

Convert to trig form. (3 points)

10. $-5\sqrt{3} - 5i$

WORK

$$r = \sqrt{75 + 25} = 10, \text{ ref } 30^\circ, \text{ QIII: } 210^\circ$$

$$\boxed{10(\cos 210^\circ + i \sin 210^\circ)}$$

WHY

Find the sum. Write it in standard form. Then graph both complex numbers and their resultant, and find the absolute value of the sum. (3 points)

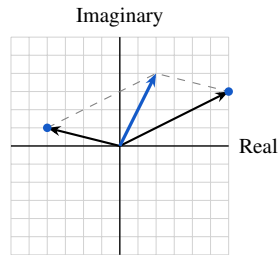
11. $(-4 + i) + (6 + 3i)$

WORK

$$\boxed{2 + 4i}$$

$$|2 + 4i| = \sqrt{4 + 16} = \sqrt{20} = \boxed{2\sqrt{5}}$$

Graph: $(-4, 1)$, $(6, 3)$, resultant $(2, 4)$



WHY

Simplify the radical.

Using $z^2 - 1$, $(z^2 - 1)^2 - 1$, $[(z^2 - 1)^2 - 1]^2 - 1$, ... is the following in the Julia set? Show ALL work. (3 points)

12. $z = -1$

YES / NO

WORK

$$(-1)^2 - 1 = 0, 0^2 - 1 = -1, \dots$$

YES

WHY

Cycles $0, -1, 0, \dots$

Perform the indicated operation. Write each answer in rectangular form. (3 points each)

13. $\frac{12(\cos 100^\circ + i \sin 100^\circ)}{4(\cos 340^\circ + i \sin 340^\circ)}$

WORK

$$3 \operatorname{cis}(-240^\circ) = 3 \operatorname{cis} 120^\circ$$

$$3 \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i \right) = \boxed{-\frac{3}{2} + \frac{3\sqrt{3}}{2}i}$$

WHY

Perform the indicated operation. Write each answer in rectangular form. (3 points each)

14. $\left[5 \operatorname{cis} \frac{5\pi}{6} \right] \left[2 \operatorname{cis} \frac{3\pi}{2} \right]$

WORK

$$10 \operatorname{cis} \left(\frac{5\pi}{6} + \frac{9\pi}{6} \right) = 10 \operatorname{cis} \frac{7\pi}{3}$$

$$\frac{7\pi}{3} - 2\pi = \frac{\pi}{3}$$

$$10 \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i \right) = \boxed{5 + 5\sqrt{3}i}$$

WHY

Using $z, z^2 + z, (z^2 + z)^2 + z, \dots$ is the following number in the Mandelbrot set? Show ALL work. (3 points)

15. $z = 1$

YES / NO

WORK

$$1, 1^2 + 1 = 2, 2^2 + 1 = 5$$

$$|5| > 2: \boxed{\text{NO}}$$

Trap: Stopping at 2 and saying yes.

WHY

2 is not over 2; keep going.

Find each power. Write your answer in rectangular form. (3 pts)

16. $(\sqrt{2} + i\sqrt{2})^{10}$

WORK

$$r = 2, \theta = 45^\circ$$

$$1024 \text{ cis } 450^\circ = 1024 \text{ cis } 90^\circ$$

$$\boxed{0 + 1024i}$$

WHY

Find all complex number solutions of each equation. Leave answers in trigonometric form. (4 pts)

17. $x^4 - 81i = 0$

WORK

$$x^4 = 81i = 81 \text{ cis } 90^\circ$$

$$3, 22.5^\circ + 90^\circ k$$

$$\boxed{3 \text{ cis } 22.5^\circ, 3 \text{ cis } 112.5^\circ, 3 \text{ cis } 202.5^\circ, 3 \text{ cis } 292.5^\circ}$$

WHY

Give two other pairs of polar coordinates for the point: one with $r > 0$ and one with $r < 0$, with $-2\pi \leq \theta < 2\pi$. (3 points)

18. $(-3, \frac{4\pi}{3})$

WORK

$$r > 0: \frac{4\pi}{3} - \pi = \frac{\pi}{3}: \boxed{(3, \frac{\pi}{3})} \text{ or } (3, -\frac{5\pi}{3})$$

$$r < 0: \frac{4\pi}{3} - 2\pi: \boxed{(-3, -\frac{2\pi}{3})}$$

WHY

Flip r , move π .

Same r , move 2π .

Convert the polar coordinates to rectangular coordinates. Write your answer in exact form. (3 points)

19. $(-8, \frac{2\pi}{3})$

WORK

$$x = -8 \left(-\frac{1}{2}\right) = 4, y = -8 \left(\frac{\sqrt{3}}{2}\right) = -4\sqrt{3}$$

$$\boxed{(4, -4\sqrt{3})}$$

WHY

Convert the rectangular equation to a polar equation. Solve for r . (3 points)

20. $x^2 + y^2 = -10y$

WORK

$$r^2 = -10r \sin \theta$$

$$r = -10 \sin \theta$$

WHY

Find the polar equation of the line that passes through the two points. Simplify your answer. (3 points)

21. $(4, 180^\circ)$ and $(2\sqrt{2}, 135^\circ)$

WORK

$$(-4, 0) \text{ and } (-2, 2)$$

$$m = \frac{2 - 0}{-2 - (-4)} = 1, \quad y = x + 4$$

$$x - y = -4$$

$$r = \frac{-4}{\cos \theta - \sin \theta}$$

WHY

Point-slope with $(-4, 0)$.

Same as $\frac{4}{\sin \theta - \cos \theta}$.

Convert each polar equation to a rectangular equation and graph it. (No calculator) (4 points each)

22. $r = \frac{6}{1 - \cos \theta}$

WORK

$$r - r \cos \theta = 6, \quad r = x + 6$$

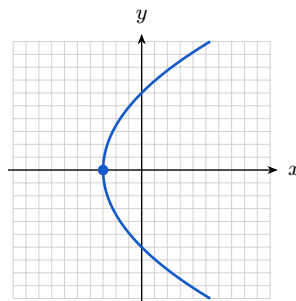
$$x^2 + y^2 = x^2 + 12x + 36$$

$$y^2 = 12(x + 3)$$

WHY

Square both sides.

Vertex $(-3, 0)$, opens right.



Convert each polar equation to a rectangular equation and graph it. (No calculator) (4 points each)

23. $r = 6 \cos \theta - 8 \sin \theta$

WORK

$$x^2 + y^2 = 6x - 8y$$

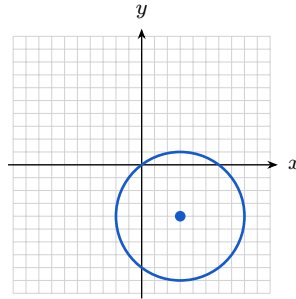
$$(x^2 - 6x + 9) + (y^2 + 8y + 16) = 25$$

$$(x - 3)^2 + (y + 4)^2 = 25$$

WHY

Complete both squares.

Center $(3, -4)$, radius 5.



Find all points of intersection for $0 \leq \theta < 2\pi$. Include the pole if the curves share it. (4 points)

24. $r = 3 \sin \theta, \quad r = 3 \cos \theta$

WORK

$$\tan \theta = 1: \theta = \frac{\pi}{4}: r = \frac{3\sqrt{2}}{2}$$

$$3 \sin \theta = 0 \text{ at } 0; \quad 3 \cos \theta = 0 \text{ at } \frac{\pi}{2}$$

$$\left(\frac{3\sqrt{2}}{2}, \frac{\pi}{4} \right) \text{ and the pole}$$

WHY

At $\frac{5\pi}{4}$ both give $r = -\frac{3\sqrt{2}}{2}$: the same point again.

Both pass through the pole.

Trap: The pole never shows up from setting $r = r$.

Identify the type of polar graph. Use the word bank: circle, cardioid, limaçon, rose curve, lemniscate. (2 points)

25. $r = 6 - 2 \cos \theta$

WORK

limaçon

WHY

$$\frac{a}{b} = 3 \neq 1.$$